DBMS

Short Notes • 10 Topic Sections

Undergraduate Computer Science Study Material

Prepared for uploading to BCS Pedia.

1. Introduction to DBMS

Data and database

Data consists of recorded facts. A database is an organized collection of related data that can be stored, searched, updated, and managed.

DBMS meaning

A Database Management System is software used to define, create, retrieve, update, and control access to databases. Examples include PostgreSQL, MySQL, Oracle Database, and SQL Server.

Advantages

A DBMS helps reduce unnecessary duplication, improve consistency, share data, enforce rules, control access, support backup and recovery, and manage concurrent users.

File system vs DBMS

Separate files can cause duplication, inconsistent values, difficult sharing, and application-dependent formats. A DBMS provides centralized management and query tools.

Users

Database administrators manage security and recovery; designers plan structures; developers build applications; end users interact with the system.

2. Architecture and Data Models

Three-schema architecture

The external level provides user-specific views; the conceptual level describes the overall logical database; the internal level describes physical storage.

Data independence

Logical data independence allows some conceptual changes without changing every user view. Physical data independence allows storage changes without changing the logical schema.

Data models

Common data models include hierarchical, network, relational, entity-relationship, and object-oriented models.

Relational model

Data is represented as relations (tables). Rows are tuples and columns are attributes. Each attribute has a domain of allowed values.

Schema and instance

A schema is the planned structure of the database. An instance is the data stored at a particular time.

3. Entity-Relationship Model

Entity

An entity is a distinguishable real-world object, such as a student, teacher, or course. An entity set contains similar entities.

Attributes

Attributes describe entities or relationships. Simple attributes are not divided further; composite attributes have parts; multivalued attributes can contain several values.

Relationships

A relationship connects entities. For example, ENROLLS can connect STUDENT and COURSE.

Cardinality

One-to-one (1:1), one-to-many (1:N), and many-to-many (M:N) describe how many entities may participate in a relationship.

Example

```
STUDENT may have StudentID, Name, and Email. A student can enroll in many courses, and a course can con
```

4. Relational Model and Keys

Relation and tuple

A relation is a table, a tuple is a row, and an attribute is a column. Attribute values should belong to appropriate domains.

Superkey and candidate key

A superkey uniquely identifies a row. A candidate key is a minimal superkey: no attribute can be removed while preserving uniqueness.

Primary key

The primary key is the candidate key selected as the main row identifier. Primary-key values must be unique and cannot be NULL.

Foreign key

A foreign key references a key in the same or another table. It helps enforce referential integrity by preventing invalid references when constraints are enforced.

Other keys

A composite key uses multiple attributes. An alternate key is a candidate key not selected as primary. A natural key has real-world meaning; a surrogate key is artificially created.

5. Relational Algebra

Meaning

Relational algebra is a procedural query language that describes operations producing relations from other relations. It provides a theoretical basis for relational query processing.

Selection and projection

Selection (σ) chooses rows satisfying a condition. Projection (π) chooses specified columns and removes duplicate tuples in classical relational algebra.

Union and difference

Union combines tuples from union-compatible relations. Difference returns tuples in the first relation but not in the second. Relations must have compatible structures.

Product and join

Cartesian product pairs every row of one relation with every row of another. A join combines related rows according to a condition.

Rename

Rename (ρ) changes a relation or attribute name. It is useful when a query refers to the same relation more than once.

6. SQL Basics and Constraints

SQL

Structured Query Language is used to define database structures, query data, modify rows, and control access. Syntax differs slightly among database products.

DDL and DML

DDL includes CREATE, ALTER, and DROP for structures. DML commonly includes INSERT, UPDATE, and DELETE for rows; SELECT retrieves rows.

Example

```
CREATE TABLE Student (
  student_id INT PRIMARY KEY,
  name VARCHAR(60) NOT NULL,
  email VARCHAR(100) UNIQUE
);
INSERT INTO Student (student_id, name)
VALUES (1, 'Asha');
SELECT * FROM Student;
```

Constraints

PRIMARY KEY identifies rows; FOREIGN KEY maintains references; NOT NULL requires a value; UNIQUE prevents duplicates; CHECK validates a condition; DEFAULT supplies a value when omitted.

Filtering

WHERE filters rows, ORDER BY sorts results, and DISTINCT removes duplicate result rows.

7. SQL Queries, Joins and Functions

Aggregate functions

COUNT counts rows or non-NULL values depending on the expression; SUM totals numbers; AVG calculates an average; MIN and MAX return the smallest and largest values.

GROUP BY and HAVING

GROUP BY forms groups for aggregate calculations. HAVING filters groups after aggregation; WHERE filters rows before grouping.

Join types

INNER JOIN returns matching rows. LEFT JOIN preserves all left-side rows and fills unmatched right-side columns with NULL. RIGHT JOIN preserves right-side rows; FULL OUTER JOIN preserves unmatched rows on both sides where supported.

Subqueries

A subquery is a query nested inside another query. It may be used in conditions, the FROM clause, or select expressions.

Example

```
SELECT department, COUNT(*) AS total
FROM Employee
GROUP BY department
HAVING COUNT(*) > 2
ORDER BY total DESC;
```

8. Normalization

Purpose

Normalization organizes tables to reduce unwanted duplication and prevent insertion, update, and deletion anomalies.

Functional dependency

A functional dependency $X \rightarrow Y$ means each value of attribute set X determines one value of attribute set Y within the relation.

First Normal Form

1NF requires atomic values under the chosen model and avoids repeating groups or lists of values in one cell.

Second Normal Form

2NF requires 1NF and full functional dependence of every non-prime attribute on each candidate key; partial dependency on part of a composite key is removed.

Third Normal Form and BCNF

3NF restricts certain transitive dependencies. BCNF requires the determinant of every non-trivial functional dependency to be a superkey.

9. Transactions, Concurrency and Recovery

Transaction

A transaction is a logical unit of work. Several statements may need to succeed together or be rolled back together.

ACID

Atomicity means all-or-nothing; Consistency preserves rules; Isolation limits interference among concurrent transactions; Durability means committed changes survive failures according to system guarantees.

COMMIT and ROLLBACK

COMMIT completes a transaction's changes. ROLLBACK cancels uncommitted changes. Exact behavior depends on transaction boundaries and the DBMS.

Concurrency control

Concurrency control coordinates simultaneous transactions to preserve correctness. Techniques include locking, timestamp ordering, and multiversion concurrency control.

Recovery

Recovery uses logs, checkpoints, backups, and other mechanisms to restore a consistent database after a crash or error.

10. Security, Indexes, Views and Revision

Security

Database security includes authentication, authorization, roles, privileges, auditing, and protection of sensitive information. Follow least privilege.

GRANT and REVOKE

GRANT gives privileges to a user or role. REVOKE removes privileges. Exact syntax varies by database system.

Indexes

An index can speed up searches, joins, and sorting. It uses storage and adds work to INSERT, UPDATE, and DELETE, so it should be used where useful.

Views

A view is a named query presented like a table. Views can simplify complex queries and restrict the rows or columns users see.

Revision questions

1. Define DBMS and list advantages. 2. Explain primary and foreign keys. 3. Differentiate DDL and DML. 4. Explain joins. 5. Describe 1NF, 2NF and 3NF. 6. Explain ACID. 7. What are indexes and views?